

# Midterm Project

*Supervised Learning - A.Y. 2024-2025 - 5-12 November 2024*

This project aims at analyzing a dataset regarding the weight loss of a pool of subjects who underwent a period of diet. In particular, we want to understand how the dataset provided information (variables) interact and influence weight fluctuations, providing a valuable resource for researchers and practitioners in nutrition and health. We want to build a system capable of predicting the *Final Weight (lbs)* of the subjects given two specific conditions:

1. Knowing in advance the subjects' Basal Metabolic Rate (BMR) at the beginning of the diet period (variable *BMR (Calories)* included in the set of available information)
2. **NOT** knowing the subjects' Basal Metabolic Rate (BMR) at the beginning of the diet period (excluding the *BMR (Calories)* variable from the set of available information).

The provided dataset includes the following information.

- `Participant ID`: Unique identifier for each participant in the study.
- `Age`: The age of the participant (in years) at the beginning of the diet period.
- `Gender`: Gender of the participant (M/F) (as physiological differences may affect weight management).
- `BMR (Calories)`: Basal Metabolic Rate, calculated using the Mifflin-St Jeor equation, representing the number of calories burned at rest.
- `Daily Calories Consumed`: Average total caloric intake per day during the diet period.
- `Daily Caloric Surplus/Deficit`: The difference between calories consumed and BMR, indicating whether the participant is in a caloric surplus or deficit (during the diet period).
- `Weight Change (lbs)`: The estimated change in weight over the specified diet duration, based on caloric surplus/deficit.
- `Duration (weeks)`: The diet time period (from 1 to 12 weeks).
- `Smoking`: Self-reported smoking habit (Yes/No)
- `Physical Activity Level`: Self-reported level of physical activity, categorized as Sedentary, Lightly Active, Moderately Active, or Very Active.

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- **Sleep Quality:** Self-reported quality of sleep, categorized as Poor, Fair, Good, or Excellent, which can affect weight management.
- **Stress Level:** A numerical score (1-10) indicating the participant's perceived stress level (stress can influence eating behaviors and weight).
- **Work Sector:** Working sector of the subject at the beginning of the diet period.
- **Final Weight (lbs):** The subject's weight at the end of the diet, serving as target for weight change.

In order to build the desired predictive system, develop the following tasks and answer the following questions. The predictive system might include more than one Linear Regression model. For all the tasks and all the choices you make, do justify your choices.

### Questions and Tasks

1. Load and explore the dataset, visualizing the results of the exploration. Analyze the quality of the dataset and handle eventual quality issues.
2. Prepare the dataset for a Linear Regression task that aims to build a "Final Weight (lbs)" predictive model.
  - (a) Verify the feature values distribution of the numerical variables.
  - (b) Perform, if necessary, features transformation for the numerical variables, justifying your choices.
  - (c) Perform, if necessary, encoding of the categorical variables, justifying your choices.
  - (d) Reason about the possibility of decreasing the dimensionality (number of features) of the dataset. Verify if different methodologies of feature selection do choose the same important features: adopt at least a univariate method, an iterative method, and a model-based method exploiting a linear regression model.
  - (e) On the base of the analysis performed above, define a dataset that you will exploit in the rest of the project (`dataset_reduced`), for the supervised task.
3. Build a supervised system based on Linear Regression capable of predicting the Final Weight (lbs) **under the two conditions specified in the preamble** (perform the following tasks for both conditions separately).  
Implement different approaches:
  - (a) Linear Regression exploiting the `sklearn` implementation of the algorithm.
  - (b) Linear Regression exploiting a **Batch Gradient Descent** learning procedure (not the *sklearn* implementation).  
Identify the best value of learning rate  $\eta$  and an appropriate number of *iterations* which ensures the convergence of the model.

- (c) Linear Regression exploiting a **Mini-Batch Gradient Descent** learning procedure (not the *sklearn* implementation), exploiting a mini-batch size of maximum 128 samples (use mini-batch sizes as power of 2).  
Identify the best value of learning rate  $\eta$  and an appropriate number of *iterations* which ensures the convergence of the model.
  - (d) Consider the complete dataset and eventually add interaction or polynomial features (`dataset_augmented`). With this dataset, train a new Linear regression model based on a **Batch Gradient Descent** learning procedure (not the *sklearn* implementation).  
Identify the best value of learning rate  $\eta$  and an appropriate number of *iterations* which ensures the convergence of the model.
  - (e) Compare the related learning curve of the Gradient Descent-based approaches developed above and comment both on the learning procedure and on the generalization performances of the model.
  - (f) With the dataset generated in the task above (`dataset_augmented`), train a Linear Regression model with Lasso regularization. Comment on the importance of each feature and reason over the number of features selected with different values of  $\alpha$ .  
Verify how the model performances change with different values of  $\alpha$  (different number of features). Use a plot of your choice.
  - (g) With the complete dataset generated above (`dataset_augmented`), train a Ridge Regression model identifying the best value of the regularization coefficient  $\alpha$  that allows the model to achieve the best generalization performances. Evaluate the generalization performances of the model.
4. Now, let's stratify the problem according to gender. Using the `dataset_reduced`, build two different supervised systems (one considering only the male subjects and one considering only the female subjects) based on Linear Regression which are capable of predicting the *Final Weight (lbs)* under the two conditions specified in the preamble. Optimize, per each system, the number of iterations of the learning procedure and the learning rate  $\eta$ . Plot the related learning curves and compare them.
  5. On this task you are going to reason on the approaches based on the `dataset_reduced`. *Do the gender-stratified approach (task 4) lead to better performances with respect to the global approach (task 3)?*  
For this task, measure how well each different approach developed above on `dataset_reduced` is capable of predicting the *Final Weight (lbs)* (under the conditions specified in the preamble) over the two genders present in the dataset. The performances to compare are the ones calculated only on test samples related to each specific gender. Visualize the obtained results.
  6. Provide a final discussion on the results obtained above.